

Scenario

A startup company named **EcoRide** is developing a **Smart E-Bike Sharing System** for large urban areas.

The system enables users to:

- Register, recharge their balance, and rent electric bikes via a mobile app.
- Locate and unlock nearby e-bikes using QR codes.
- Be charged based on time and distance.
- Allow the control center to monitor battery levels and maintenance schedules.

The company plans to apply the **COMET (Collaborative Object Modeling and Architectural Design Method)** and **UML 2** to design an initial software architecture that ensures **scalability**, **maintainability**, and **integration** with third-party services (e.g., maps, e-payment gateways, IoT-based tracking).

Question 1: Use Case Modeling (25%)

Identify the **main actors** and at least **three key use cases** for the EcoRide system.

- Draw a **Use Case Diagram** showing the relationships between actors and use cases (use include or extend relationships where appropriate).
- Briefly describe each actor's role and the goal of each use case.

Question 2: Static Modeling (25%)

Based on your use cases:

- Identify the **main classes** and classify them as **Entity**, **Boundary**, or **Control** classes according to COMET guidelines.
- Draw a **Class Diagram** showing **associations**, **aggregations/compositions**, and **generalization/specialization** relationships if applicable.
- Explain how these classes contribute to realizing the system's functionality.

Question 3: Dynamic Interaction Modeling (25%)

Choose one key use case (e.g., *Rent an E-Bike*) and describe the **sequence of object interactions** when executing that use case.

- Draw a **Sequence Diagram** or **Communication Diagram** to illustrate message exchanges between objects.
- Describe the **main flow** and at least **one alternative flow** (e.g., bike unavailable, low balance, or connection error).

Question 4: Architectural Design (25%)

Propose a **software architecture** for the EcoRide system using **one of the following styles**:

- *Client/Server Architecture*
- *Service-Oriented Architecture (SOA)*
- *Component-Based Architecture*

Explain:

1. Why this architectural style was chosen.
2. The **major subsystems/components** and how they communicate.
3. The **software quality attributes** your design supports (e.g., scalability, modifiability, performance, security).